

20241105EB_MN (TDP43 w/ Thiamet-G and Ver, NCT with rescue)

Project: Yi-Shuan Tseng Holt Lab
Author: Yi Shuan Tseng
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- 2024年11月5日 星期二
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2024年11月7日 星期四
- T2 -> T3
- 2024年11月9日 星期六
- T3 -> T4 (~14 mL for over-weekend)
- 2024年11月11日 星期一
- T4 -> T5
- 2024年11月13日 星期三
- T5 -> T6
- 2024年11月15日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2024年11月18日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 2. Remove the PLO and wash the plate with PBS once
 3. Coat the plate with laminine for 1hr.
 4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 5. Add trypsin (usually 3 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
 6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 6mL of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 6mL of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

conc.: 570K/mL(100%)

total 2.85M

-> 14 wells of 24-well plate (204K/well)

Table1			
	A	B	C
1	A	4 wells	no virus
2	B	5 wells	110 ~12-14uL
3	C	5 wells	111 10uL

Well1						
	1	2	3	4	5	6
A		A1			C1	
B		A2	B1	C4	C2	
C		A3	B2	B4	C3	C5
D		A4	B3	B5		

2024年11月19日 星期二

Change T7 -> T7 if added virus (not totally sure if I changed media here. I did not write it on the notebook)

2024年11月20日 星期三

T7-> BP T8

2024年11月22日 星期五

T8 -> T8 (all, because I accidentally added Hypo media into the well for 3min. I removed all media and added 500uL T8 into each well)

2024年11月25日 星期一

200 uL BP T8 -> 300 uL BP T8

2024年11月27日 星期三

Image set A

media is prepared with 2.5mM JFX650 and SPY595 1:2500

O.C. , 60X, 704x704, 16-bit					^
	A	B	C	D	
1	channel	bit	power (%)	exposure time	
2	561	16	10	200 ms	
3	640	16	20	100 ms	

Imaging: 12 postions/well

Table2									
	A	B	C	D	E	F	G	H	I
1	number	condition	media volume (mL)	JFX/SPY volume (uL)	drug volume	treatment time	image time (1h)	image time (3h)	image time (24h)
2	1	2.5mM (BP)	0.8	0.32	NA	3:25	16:45	~18:00-18:30	~18:00-18:45 on 11/28
3	2	0.15mM (hypo)	2.25	0.9	NA	3:55	16:55		
4	3	0.15mM+Ver1 50			2.25	4:25	17:45		
5	4	0.15mM + Thiamet 10uM			0.75	4:55	17:55		

2024年11月28日 星期四

Image set B&C
add 300uL/well

O.C. , 60X, 600x900, 12-bit					^
	A	B	C	D	
1	channel	bit	power (%)	exposure time	
2	640	12	15	100 ms	
3	561	12	20	200 ms	
4	488	12	50	1s	

Table3																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1			110 (B)									111 (C)						
2	number	condition	BP/hypo treatment time	rescue time	image ID	image 1h	image ID	image 3h	image ID	image 24h	BP/hypo treatment time	rescue time	image ID	image 1h	image ID	image 3h	image ID	image 24h
3	1	2.5mM (BP)	9:00	x	1	10:10	7	11:40	33	11/29 9:55	12:35	x	15	13:38	23	15:01	39	11/29 11:26
4	2	0.15mM (hypo)	9:30	x	3	10:36	11	12:37	35	11/29 10:27	13:05	x	17	14:07	25	16:04	41	11/29 12:00
5	3	0.15mM+1h rescue	10:05	11:00	9	12:06		x		x	13:05	14:05	21	15:01		x		x
6	4	0.15mM+24h rescue	11/27 15:00	11/27 16:00 (11/28 14:05 change)	29	17:04		x		x	11/27 15:00	11/27 16:00 (11/28 change 14:05)	31	17:41		x		x
7	5	0.15mM + Thiamet 10uM	10:05	x	5	11:01	13	13:08	37	11/29 10:55	13:36	x	19	14:32	27	16:31	43	11/29 12:27

2024年11月29日 星期五

Take 24h time point of set B&C

** cells are not very healthy (maybe the media is not enough since I only added 300uL yesterday?

** DO NOT use the data at 24h timepoint